9 batterier $\quad \bar{X}$ = pakker tilfeldig batteri

3 = A ∩ } uavhengige
6 = B ∩ }

$E[A] = 12$ timer $= \frac{1}{\lambda} \Rightarrow f_A(a) \ \frac{1}{12} e^{-\frac{1}{12}a}$

$E[B] = 20$ timer

B ∩ A ~ exp

$P(\bar{X} > 15) = 1 - P(\bar{X} \leq 15)$

$P(\bar{X} > 15) = P(X > 15 | A)P(A) + P(X > 15 | B)P(B)$

$1 - P(\bar{X} \leq 15) = 1 - [P(A \leq 15)P(A) + P(B \leq 15)P(B)]$

$= 1 - \left[F_A(15) \frac{3}{9} + F_B(15) \frac{6}{9} \right]$

$= 1 - \left[\left(1 - e^{-\frac{1}{12} \cdot 15}\right) \frac{1}{3} + \left(1 - e^{-\frac{1}{20} 15}\right) \frac{2}{3} \right]$

$= 1 - \left[\frac{1}{3} - \frac{1}{3} e^{-5/3} + \frac{2}{3} - \frac{2}{3} e^{-3/4} \right]$

$= \frac{1}{3} e^{-5/3} + \frac{2}{3} e^{-3/4}$

$= 0{,}4104$

b) $P(X = B \mid \bar{X} > 15) =$ (Bayes regel) $\frac{P(\bar{X} = B \cap \bar{X} > 15)}{P(\bar{X} > 15)}$

$= \frac{P(\bar{X} > 15 | X = B) P(B)}{0{,}4104} = \frac{e^{-3/4} \cdot \frac{2}{3}}{0{,}4104} = 0{,}7674$

Opg 2

Alle celler må virke for at pack funker

72 solceller

Levetid $\sim exp(\lambda)$ $\Big\}$ $f_L(l) = \frac{1}{5000} e^{-\frac{1}{5000} l}$

$E[L] = 5000$

a) Tilfældig valgt celle $= X$

$$P(L > 25) = 1 - F_L(25) = e^{-\frac{25}{5000}} = 0{,}9950$$

b) $$P(L > 30 \mid L > 5) = \frac{P(L > 30 \cap L > 5)}{P(L > 5)} = \frac{P(L > 30)}{P(L > 5)}$$

$$= \frac{1 - F_L(30)}{1 - F_L(5)} = \frac{e^{-\frac{30}{5000}}}{e^{-\frac{1}{1000}}} = 0{,}9950$$

c) $P(72 \text{ celler med } L > 25) = P(L_1 > 25 \cap L_2 > 25 \cap \dots L_{72} > 25)$

$$= P(L > 25)^{72} = 0{,}6977$$

Oppg 3

Levetid til pakning: $T_1, T_2, T_3 \sim \exp(\lambda = \frac{1}{2})$

Y = tid til 3. pakning svikter $\Big\} Y \sim \text{gamma}(\alpha, \beta)$

$\alpha = 3 \quad \beta = \frac{1}{\lambda} = 2$

a) $E[Y] = \alpha\beta = 6$

b) $\text{Var}[Y] = \alpha \cdot \beta^2 = 12$

c) $P(\text{minst 1 funksjonde pakning etter 5 år})$

$= 1 - P(\text{Alle sviktet pakning før 5 år})$

$= 1 - P(Y \leq 5)$

$= 1 - \int_0^5 \frac{1}{\beta^\alpha \Gamma(\alpha)} t^{\alpha-1} e^{-\frac{t}{\beta}} dt, \quad \Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$

$\Gamma(3) = 2$

$= 1 - \frac{1}{16} \int_0^5 t^2 e^{-\frac{t}{2}} dt = \underline{0{,}5438}$

$1 - \frac{1}{16}\left[t^2 \cdot (-2) e^{-\frac{t}{2}} + 4 \int t e^{-\frac{t}{2}} dt \right]$

$$= 1 - \frac{1}{16}\left[t^2 \cdot (-2) e^{-\frac{t}{2}} + 4\left[t \cdot (-2) e^{-t/2} + 2 \int_0^{-t/2} dt \right] \right]_0^5$$

$$= 1 - \frac{1}{16}\left[t^2 \cdot (-2) e^{-\frac{t}{2}} + 4\left[t \cdot (-2) e^{-t/2} + 2 \int_0^{-t/2} dt \right] \right]_0^5$$

$$= 1 - \frac{1}{16}\left[-2t^2 e^{-t/2} + 4\left[-2t e^{-t/2} - 4 e^{-t/2} \right] \right]_0^5$$

$$= 1 - \frac{1}{16}\left[-50 e^{-5/2} - 40 e^{-5/2} - 16 e^{-5/2} \right] - \left(-\frac{1}{16} \cdot -16 \right) = \underline{0,544}$$

Opg 4)

$$T = 2\pi\sqrt{\frac{L}{G}}$$

a) $G: \left(\frac{T}{2\pi}\right)^{-2} \cdot L$, $G = h(L,T)$

b) $\mu_L = 10$, $\sigma_L = 0,03$

$\mu_T = 6,34$, $\sigma_T = 0,44$

$\frac{\partial h}{\partial L} = \left(\frac{T}{2\pi}\right)^{-2}$

c) $\frac{\partial h}{\partial L}$, $T = \mu_T : \left(\frac{6,34}{2\pi}\right)^{-2}$

d) $\frac{\partial h}{\partial T} = \frac{\partial}{\partial T}(2\pi)^2 \cdot L \cdot \frac{1}{T^2} = (2\pi)^2 \cdot L \frac{\partial}{\partial T} \frac{1}{T^2} = (2\pi)^2 L \left(-2 \cdot \frac{1}{T^3}\right)$

f) $\sigma_G = ?$ $Var[G] = \left[\frac{\partial h}{\partial L}(\mu_L, \mu_T)\right]^2 \cdot Var[L] + \left[\frac{\partial h}{\partial T}(\mu_L, \mu_T)\right]^2 \cdot Var[T]$

$= 0,982^2 \cdot 0,03^2 + (-3,098)^2 \cdot 0,44^2$

$= 1,852$

$\sigma_6 = \sqrt{1,852} = 1,363$

Opg 5

$\bar{X} \sim N(\mu, \sigma^2)$, $Y = e^X$

c) $f_Y(y) = ?$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \frac{1}{\sigma} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}$$

$y = u(x) = e^x$ $\quad \Rightarrow w(y) = x(u)$, $w'(y) = \frac{d}{dy} \ln y$

$x(u) = \ln|u|$ $\quad w'(y) = \frac{1}{y}$

$$f_Y = f_X(w(y)) \cdot \left|\frac{1}{y}\right| = \frac{1}{\sqrt{2\pi}} \frac{1}{\sigma} e^{-\frac{1}{2}\frac{((\ln y)-\mu)^2}{\sigma^2}} \cdot \frac{1}{y}$$

b) $Var[Y] = Var[e^X] \approx (e^\mu)^2 \cdot Var[X] = e^{2\mu}\sigma^2$

$\frac{d}{dx} e^x = e^x$

c) $\varepsilon = \left| \frac{e^2 \sigma_1^2 - e^{2+\sigma_1^2} \cdot (e^{\sigma_1^2} - 1)}{e^{2+\sigma_1^2} \cdot (e^{\sigma_1^2} - 1)} \right|$

d) $\sigma = 0.6$

Pag 6/ $f_X(x) = \begin{cases} 2x^2 \phi(x) & x \geq 0 \\ 0 & x < 0 \end{cases}$

$\phi(x) =$ standard $\sim N$ $\qquad \phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$

$Y = aX^b$

$f_Y(y)$: $\qquad w(y) = \sqrt[b]{\frac{y}{a}}$

$w'(y) = \left(a^{-\frac{1}{b}} \cdot y^{\frac{1}{b}}\right)' = \frac{1}{\sqrt[b]{a}} \cdot \frac{1}{b} \cdot \frac{1}{\left(\sqrt[b]{y}\right)^{b-1}}$

$f_Y(y) = f_X(w(y)) \cdot |w'(y)|$

$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\sqrt[b]{\frac{y}{a}}\right)^2} \cdot 2\left(\sqrt[b]{\frac{y}{a}}\right)^2 \cdot a^{-\frac{1}{b}} \cdot \frac{1}{b} \cdot \frac{1}{\left(\sqrt[b]{y}\right)^{b-1}}$

$\frac{2}{\sqrt{2}} = \frac{\sqrt{2}}{1}$

$\frac{1}{b} a^{-\frac{2}{b}} \cdot a^{-\frac{1}{b}} \cdot y^{-\frac{b-1}{b}} \cdot y^{\frac{2}{b}}$

$\frac{1}{b} \cdot a^{-\frac{3}{b}} \cdot y^{\frac{3}{b}-1}$

$$a^{-3/b} = \frac{a^{3/b-1}}{a}$$

$$f_Y(y) = \frac{\sqrt{2}\cdot\left(\frac{y}{a}\right)^{3/b-1}\cdot e^{-\frac{1}{2}\left(b\sqrt{\frac{y}{a}}\right)^2}}{\sqrt{\pi}\,a\,b}$$

b) $a \wedge b = ?$ $\Rightarrow$ $Y =$ khi_kvadrat fordelt

Def. Kjikvadratfordelt:

$$X = Z_1^2 + Z_2^2 + \cdots \sum_{i=1}^{\nu} Z_i^2 \qquad Y = aX^b$$

$Z \sim \phi(x)$

$a = 1 \wedge \beta = 2 \qquad Y = X^2$

frihetsgrader $= \nu$ $\Rightarrow$ $y \sim$ gamma (ν)

Kjikvadrat: $f_y = \frac{1}{2^{\nu/2}\,\Gamma\left(\frac{\nu}{2}\right)}\, y^{\frac{\nu}{2}-1}\, e^{-\frac{x}{2}}$

Vår ferd: $f_Y = \frac{1}{ab\sqrt{\pi}}\left(\frac{y}{a}\right)^{\frac{1}{b}-1} e^{-\frac{1}{2}\left(a\sqrt{\frac{y}{a}}\right)^2}$

$c=1, b=2$: $\frac{1}{2^{\frac{3}{2}}\Gamma(\frac{3}{2})} y^{\frac{3}{2}-1} e^{-\frac{y}{2}}$

$y^{\frac{3}{2}-1} = x^{\frac{\nu}{2}-1} \Rightarrow \underline{\nu = 3}$

$2^{3/2} \cdot \Gamma(\frac{3}{2}) = \sqrt{8} \cdot \frac{\sqrt{\pi}}{2} = 2\sqrt{2}\frac{\sqrt{\pi}}{2} = \underline{2\sqrt{\pi}}$

Q.E.D